How Much Fault Tolerance Is in a Fly Brain?

Fault-Injection Analysis of a Whole-Connectome Spiking Controller for Physical Intelligence

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***Abstract*—Whole-brain point-neuron models built from complete synaptic wiring diagrams have begun to serve as controllers in embodied settings, but their behavior under component-level faults has never been measured. We take the released leaky integrate-and-fire (LIF) whole-brain model of *Drosophila melanogaster* (138,639 neurons, 15.1M signed connections, 54.5M synapse contacts) and inject five hardware-motivated fault classes—weight variability, synapse loss, random unit loss, hub loss, and sensory-channel deafferentation—scoring behavioral outcome on descending command neurons. Across 36 fault conditions (12 seeded trials each), the connectome absorbs 50% lognormal weight variability, retains 76% of escape-command activity after 50% synapse loss, and shows no median change under 10% random unit loss, while loss of the top-1% out-degree hubs produces a 2.5× runaway of global activity rather than silence. An edge-permuted null preserving weight and degree statistics transmits nothing, and relative pathway decay under edge loss is topology-independent: the observed robustness is consistent with threshold amplification on convergence structure, not with graph redundancy alone. We distill the results into five design principles for fault-tolerant neuromorphic controllers.**

***Index Terms*—neuromorphic computing, connectome, fault tolerance, fault injection, Drosophila**

I. Introduction

PHYSICAL intelligence—agents whose perception, decision, and action loops close in the physical world under tight energy and latency budgets—demands controllers that remain robust on imperfect hardware [1], [2]. Neuromorphic systems promise the required efficiency [3], [4], and spiking implementations on Loihi 2 and FPGA substrates now cover vision and control workloads [5]–[7]. What remains poorly quantified is how *wiring* determines robustness: engineered spiking networks are designed task-by-task, so their fault tolerance must be re-measured per design [8], [9]. Complete, signed wiring diagrams of whole brains are now public, and point-neuron models built on them reproduce realistic dynamics without training [10], [11]; an evolved connectome is thus a field-tested controller and a reference point for engineered fault tolerance.

We perform, to our knowledge, the first systematic *fault-injection* study on such a controller: the released leaky integrate-and-fire (LIF) whole-brain model of *Drosophila melanogaster* [10][1], which instantiates the complete FlyWire connectome [11]. We instrument the visuomotor escape pathway: looming-sensitive visual interneurons (LC4, LPLC2) converge onto descending command neurons (DNp01, DNa02, DNp04, DNb01) whose activation evokes take-off and escape steering [12]; outcome is scored as command-population firing rate. Its failure modes map directly onto hardware fault classes: weight imprecision, broken interconnect, dead cores, dead sensors.

We inject five fault families (Section III-C), each over 12 seeded 600 ms trials, with two controls: an edge-permuted null preserving weights and degree statistics, and an analytic multi-hop pathway-mass computation. Our contributions: (i) a fault-injection framework mapping hardware failure modes onto a biologically exact 138k-neuron controller; and (ii) the first behavioral-grade tolerance curves of a complete connectome (50% weight noise absorbed, 50% synapse loss retains 76%, 10% unit loss leaves the median unchanged): top-1% hub loss instead produces a 2.5× *activity runaway* that passes transmission while violating the output envelope, and whole-channel deafferentation is the only catastrophic mode. We further (iii) decompose why graph redundancy alone explains neither specificity nor graceful degradation, offering threshold amplification as a candidate account, and (iv) distill five design principles for fault-tolerant neuromorphic controllers.

II. Related Work

The release of complete wiring diagrams for the adult *Drosophila* brain [11] has shifted connectome research from static graph analysis to executable models spanning visual system convolutions [13], neuron-level response prediction [14], learned tasks and whole-body locomotion on whole-CNS wiring [15]–[17], and biofidelic processing units [18], with cell-type studies resolving descending-neuron input sharing [19], [20]. Closest to our substrate, the released LIF whole-brain model (footnote 1) reproduces global activity and has been ported to Loihi 2 [5]. None of these works varies the integrity of the controller; we treat the connectome controller as a fixed plant and sweep internal fault modes.

Graph-theoretic lesion studies score the structural attack tolerance of brain networks [21], not dynamics; spiking-network weight-noise robustness is mostly characterized on small trained architectures [8], [9], with adaptive mitigation now studied at the edge [22]; purpose-built fault-injection frameworks for trained SNNs—SpikeFI [23], SpikingJET [24]—automate this but score task accuracy on human-designed networks, continued by STDP campaigns and population-coded redundancy [25], [26]. Hardware platforms expose complementary constraints: on-chip variability co-design [6],

[1]https://github.com/philshiu/Drosophila_brain_model

FPGA precision–capacity trade-offs [7], permanent interconnect faults [27], edge energy–robustness envelopes [3], [28], and trained-CNN dependability [29]. Benchmarking efforts omit fault-injection axes [30], and formalizations of neuromorphic control do not yet treat degradation [31]. Insect-inspired collision-aversion circuits are robust by construction, but small and hand-designed [32], [33]. What a complete connectome uniquely enables is tolerance measurement at the scale of an entire evolved controller; the curves read as requirements rather than observations.

III. Fault-Injection Framework

A. Whole-Brain Spiking Substrate

We adopt the released whole-brain LIF model of *Drosophila* [10] (Section I), implemented in Brian2 [34] and built on the FlyWire connectome [11]. Every neuron is a current-based LIF unit (threshold -45 mV, reset -52 mV, membrane time constant 20 ms, refractory 2.2 ms, axonal delay 1.8 ms). Each connection in the released table becomes a delta-conductance synapse whose jump is 0.275 mV per counted contact and decays with a 5 ms synaptic time constant, signed by the presynaptic cell's neurotransmitter; the connectivity table yields 15,091,983 connections between 138,639 neurons (54.5M contacts in total), 40% of connections inhibitory, with global activity sustained by the released model's calibrated Poisson background drive. Our engine restores the full network state at the start of every trial, so all conditions share identical initial states, while the Poisson stimulus and background draw fresh realizations each trial: reported dispersion includes stimulus variability. One 600 ms trial runs in $\approx$3.5 s on one CPU core.

B. Behavioral Task: Visuomotor Escape

The instrumented pathway (Fig. 1) is the fly's looming-triggered escape circuit: looming-sensitive visual projection neurons LC4 (104) and LPLC2 (210) converge on descending neurons [12], [32]. Because collision-sensing models treat the looming response as sustained high-rate drive, we drive the sensory cells—and, as a control, an equal-sized random neuron set—with independent 150 Hz Poisson trains gated per trial, rather than modeling looming kinetics. The behavioral readout is the population rate of the command set {DNp01, DNa02, DNp04, DNb01} (8 neurons) over the full 600 ms trial; all 1,299 descending neurons serve as a secondary readout. Calibration drove the command population to 94.6 Hz (interdecile 93.6–95.8 Hz, first-spike latency 4.4 ms; 10-trial set) under escape stimulation but only 8.2 Hz (6.4–9.2 Hz) under random stimulation; $\theta = 51.4$ Hz (the 94.6/8.2 Hz midpoint), and every pass/fail classification below is unchanged for any threshold between the observed trial extremes, 9.2 and 63.1 Hz; Section IV baselines use the 12 sweep trials.

C. Fault Taxonomy

Five fault families cover the hardware-relevant failure modes. F1 multiplies every synaptic weight by an i.i.d. mean-preserving lognormal factor ($\sigma \in \{0.05, 0.1, 0.2, 0.5\}$), modeling analog weight variability and bit-level imprecision.

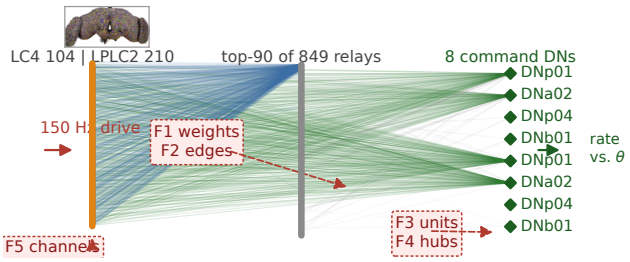


Fig. 1. The instrumented escape pathway, drawn from the connectome itself. Left: the 314 driven sensory cells (LC4, LPLC2); center: the 90 top-ranked of 849 two-hop relays; right: the 8 command DNs. Line width and opacity scale with synapse count (594 direct sensory→DN edges); dashed red marks the fault-injection points (F5 channels; F1/F2 synapses; F3/F4 units). Inset: reconstructed adult brain, adapted from Dorkenwald et al. [35] (CC BY 4.0).

F2 deletes a random fraction $f \in \{0.05, 0.10, 0.25, 0.50\}$ of synapses, modeling interconnect or memristor failures. F3 silences all output synapses of a random fraction $f \in \{0.01, 0.02, 0.05, 0.10\}$ of units, excluding input and readout cells (unit removal in this delta-synapse model), modeling dead cores. F4 applies the same operation to the top 1%, 2%, or 5% of units ranked by total outgoing synapse count (again excluding input and readout), modeling failure of high-fanout components; these cells carry 9–22% of all connections (1%, 2% and 5% sets, respectively), a targeted, correlated version of F3. F5 zeroes all outgoing synapses of one (LC4 or LPLC2) or both sensory channels, modeling sensor or afferent-path failure. All faults are applied after state restoration; F4 and F5 act on fixed cell sets, while F1–F3 redraw their realization each trial from SHA-256 seeds over (phase, condition, trial)—12 independent, exactly reproducible draws per condition.[2]

D. Controls

Two controls isolate what the biological wiring contributes. The first is an edge-permuted null: every synapse keeps its presynaptic partner, weight, and sign, but its postsynaptic target is randomly reassigned. This preserves out-degrees, weights, and the global in-degree histogram while destroying which inputs converge onto each target—and with them all multi-hop structure; the identical sweep runs on it. The second is analytic: for both graphs we compute the *pathway mass* $m_k = \mathbf{1}_{\mathrm{DN}}^{\top} A^k \mathbf{1}_{\mathrm{in}}$, the synapse-weighted total of all k-hop directed paths from LC4∪LPLC2 into the full descending layer (A holds contact counts, i.e., absolute signed strengths; m_k counts k-step walks of either sign—a wiring-density proxy, not signal flow), on the intact graph and after the F2 deletion levels; comparing its decay with the behavioral decay separates topological redundancy from dynamical amplification.

IV. Results

A. Baseline: The Connectome Is a Specific, Low-Latency Controller

Driving LC4 and LPLC2 at 150 Hz produces a command-population rate of 94.7 Hz (median over 12 trials; interdecile

[2]Code, connectome-derived edge lists, and per-trial records: https://github.com/Scophield/drosophila-fault-injection.

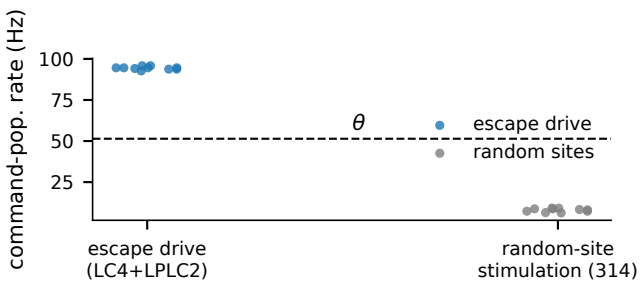


Fig. 2. Readout specificity. Command-population rate under escape stimulation (LC4+LPLC2) versus matched random-site stimulation (314 cells); dashed line: θ.

94.0–95.6 Hz) with 4.05 ms median first-spike latency, far above θ (Fig. 2). An equal-sized random set, driven identically, yields only 8.2 Hz on the same readout while producing *more* total activity (293k vs. 40k spikes): the readout is pathway-specific, not generic arousal. When both sensory channels are silenced ($F5_{\text{both}}$), the command rate collapses to exactly zero in every trial while the rest of the brain keeps spiking at background level: the score measures transmission through the instrumented pathway.

B. Tolerance Curves: Noise Is Absorbed, Disconnection Degrades Gracefully

Figure 3 shows dose–response curves for the two most consequential families; the F1 and F3 responses are flat in median (Table I). Weight variability (F1) leaves the median unchanged: at σ=0.5 the rate is 94.2 Hz (healthy: 94.7 Hz) while the trial dispersion widens (interdecile 90.4–100.8 vs. 94.0–95.6 Hz healthy), and total activity rises only from 39.7k to 41.6k spikes. Random synapse deletion (F2) degrades gracefully: 5% and 10% loss leave the rate at 91.9 and 93.1 Hz, 25% loss retains 84.5 Hz (89.2% of baseline), and even 50% loss retains 72.4 Hz (76.5%), with all 12 trials passing the transmission criterion in every condition (weakest trials 63.1 and 68.1 Hz—above θ by $\geq$22%). Disconnection's cost appears predominantly in latency: 4.05 to 5.4 ms at 50% loss (+33%), consistent with recruitment of longer residual multi-hop routes; which dominates a task-level error budget is workload-dependent. Random unit loss (F3) up to 10%—nearly 14,000 neurons with all their synapses—leaves the median unchanged (95.1 vs. 94.7 Hz): command drive integrates over hundreds of converging populations, and no tested random 10% slice is load-bearing. The wide 1,299-neuron descending layer tracks the same pattern (2,941 spikes healthy vs. 1,918 at 50% loss), so the graded response is a property of command encoding, not of the eight-neuron readout.

C. Hub Loss Produces Runaway, Not Silence

The dose–response of hub removal (F4) is non-monotonic and qualitatively different from every distributed fault (Figs. 3b, 4b). Removing the top 1% of cells ranked by total outgoing synapse count (1,386 neurons, 9.1% of all connections) *increases* the command rate to 122.3 Hz (+29%, interdecile 121.1–123.9 Hz across trials) and more than doubles global activity (99.9k spikes, 2.52× healthy; Mann–Whitney vs. healthy $p < 10^{-4}$, as for $F2_{50}$ and $F5_{\text{both}}$); at 2% the rate stays elevated (117.1 Hz) with activity partially normalized (42.4k); at 5% (22.4% of connections) both metrics return to near-baseline (94.2 Hz, 36.4k). A matched-scale contrast shows this is not a size effect: random 10% unit ablation disconnects 10.0% of connections (more than the hub set's 9.1%) yet leaves the readout unchanged (95.1 Hz). Hub lesions do not starve the pathway of drive—they remove broadly connected inhibitory regulation, and spared circuitry over-drives the command layer: about 62% of removed hub output connections are inhibitory, versus 40% network-wide. We therefore score *transmission* (rate $\geq \theta$) and an *activity envelope* (rate and total activity within 2× of healthy) separately: hub loss at 1% passes transmission in all trials while violating the envelope in all 12 (the violation persists for any envelope multiplier between 1.5× and 2×); every distributed fault passes both; $F5_{\text{both}}$ fails both. The envelope violation is the engineered system's dangerous mode: output exceeds its envelope while every per-unit monitor reads "active." The same lesion in the permuted null produces no such signature, tying the runaway to biological hub structure, not degree heterogeneity per se.

D. What the Wiring Contributes: Null Network and Pathway Analysis

The permuted null dissociates specificity from redundancy: out-degrees, weights, and the in-degree histogram survive, convergence is reassigned, and the null transmits *nothing* to the command layer (0.0 Hz in every condition and trial); its broad 1,299-neuron readout falls to background (45.5 spikes per trial vs. 2,941 healthy) and global activity to about 60% of the biological level (23.8k vs. 39.7k)—it spikes, but nothing converges onto descending output. Pathway mass quantifies the convergence advantage: 5.8× more 1-hop and 17.7× more 4-hop route mass from sensory input into the descending layer than the null. Yet *relative* pathway-mass decay under the same deletion levels is nearly identical in both graphs (at f=0.5, k=8: 0.3% vs. 0.4% of baseline; Fig. 4a). Combined with behavior degrading far more slowly than pathway mass (76% functional retention at f=0.5 against 5.8% four-hop route mass), this rules out graph redundancy alone and is consistent with a threshold-amplification account of the topological-to-functional conversion. The comparison does not isolate the dynamics by itself: the 594 direct sensory→DN connections, which survive edge deletion in proportion, and excitation–inhibition rebalancing could contribute; targeted controls are needed. A threshold-amplification account is simple: with N afferents converging and per-synapse survival p, converged drive scales as Np while k-hop route mass decays as p^k (6.25% at k=4, p=0.5); the threshold's saturating nonlinearity maps drive onto output, so the rate retains 76.5% at p=0.5 and 89.2% at p=0.75, above the linear 50% and 75% bounds the account predicts; near-threshold drive is where it should

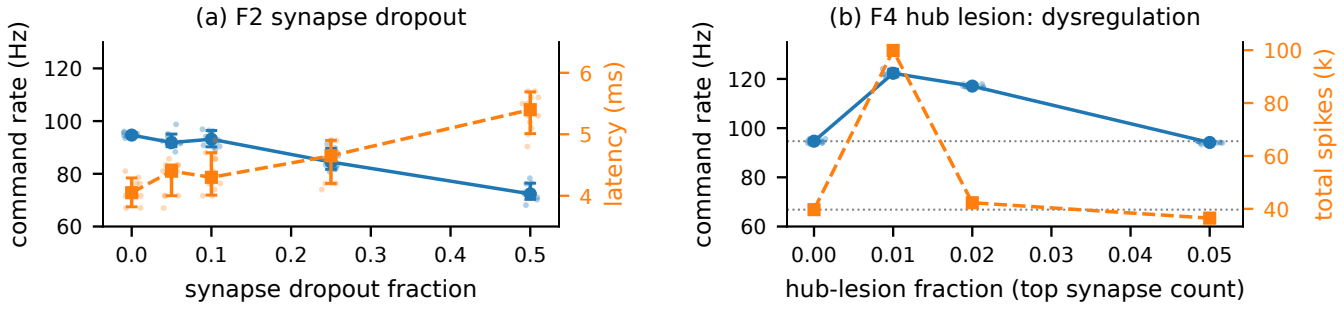


Fig. 3. Behavioral tolerance curves (medians over 12 trials, blue: rate; orange: latency ms (a) / total spikes (b); dots: trials; dotted grey: healthy medians). (a) Synapse dropout: 76% of the command rate retained at 50% loss while latency rises 33%. (b) Hub lesion: dysregulation, not signal loss.

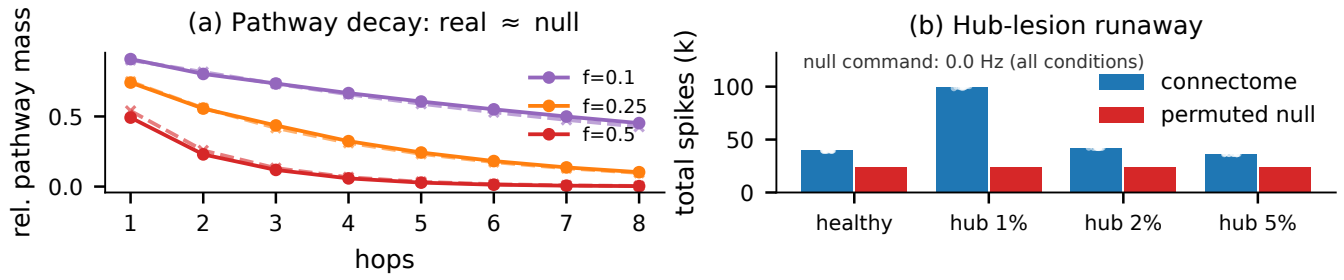


Fig. 4. Topology contrast. (a) Route mass under edge loss, normalized per graph (solid: connectome, dashed: permuted null): relative decay nearly identical, absolute mass 5.8–17.7× higher. (b) Activity under hub lesions: runaway only in biological wiring; null command readout at 0.0 Hz.

TABLE I
DESIGN PRINCIPLES FROM CONNECTOME FAULT INJECTION, WITH QUANTITATIVE ANCHORS.

#	Principle	Anchor (this study)
P1	Buy robustness with redundancy of wiring, not precision of weights	σ=0.5 lognormal noise: 99.4% retention
P2	Distribute command integration; no tested slice load-bearing	10% unit loss: 100.4% retention
P3	Budget latency, not accuracy, for disconnection faults	50% synapse loss: 76% rate, +33% latency
P4	Regulate high-fanout units; monitor envelope, not liveness	top-1% hub loss: 2.5× activity runaway
P5	Duplicate the sensory interface before the network	single-channel loss 68–79% median; both channels 0%

fail, saturation compressing increases more than decreases. Graceful degradation here reflects convergent wiring preserving aggregate drive plus threshold dynamics compressing drive changes into small rate changes.

V. Design Principles for Neuromorphic Embodied Controllers

Table I distills five transferable principles. P1 and P2 follow from the flat F1 and F3 medians: computation lives in distributed convergence, not precise magnitudes, favoring dense low-precision, partially defective fabrics—consistent with hardware observations [6], [8]. P3 shifts the failure budget: disconnection costs are mostly latency here, so real-time loops should specify worst-case response time under edge loss, which Fig. 3a reports directly. P4 is the most actionable: the only *amplifying* failure mode comes from removing the highest-fanout units, so designs should cap high-degree influence and monitor output-envelope violation rather than liveness—over-activity while all units read "active." P5 locates the single point of failure at the interface: each sensory channel alone suffices (68–79%); front-end duplication buys more safety than any internal redundancy.

One methodological warning: identical relative pathway decay in both graphs, yet only the biological one transmitting a decodable command, warns against certifying embodied controllers by topology alone; fault-injection campaigns belong in pre-deployment stages [30], [31], and the seed-exact sweep re-runs in minutes on other hardware mappings [5].

VI. Limitations and Conclusion

Our measurements inherit the LIF abstraction—single-compartment delta synapses, no plasticity, Poisson drive—and a single drive rate; doses kept the background regime intact, the hub analysis is descriptive, and the taxonomy excludes correlated damage and recovery. An evolved 138k-neuron controller keeps its median command rate under weight noise and unit death and three quarters of it at half its connections gone—given convergent wiring and one live sensory channel; its dangerous failure is hub-loss dysregulation: robustness lives in the wiring, not in precision parts.

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